

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250275649

Kind Code

A1

Publication Date

September 04, 2025

Inventor(s)

Kidd, IV; James Mallory

BBQ VENTING SYSTEM

Abstract

A BBQ grill venting system is disclosed that allows for the customization of different cooking zones within a single BBQ cooking grill. The grill includes a plurality of adjustable vents that permit the precise tuning of venting conditions within the heated interior of the grill cooking area under the BBQ lid. The variable vents are arranged in a circumferential pattern around the lower rounded top surface of the BBQ lid so that grouping of controlled temperature zones and internal atmospheric adjustments may be made to the interior of a selected cooking zone from the exterior adjustment of the different venting groups. The adjustments allow for the controlled cooking of different types of meats in different zones on a single cooking grate instance in the BBQ grill.

Inventors: Kidd, IV; James Mallory (Birmingham, AL)

Applicant: Kidd, IV; James Mallory (Birmingham, AL)

Family ID: 96881350

Appl. No.: 19/066081

Filed: February 27, 2025

Related U.S. Application Data

us-provisional-application US 63560668 20240302

Publication Classification

Int. Cl.: A47J37/07 (20060101); A23L5/10 (20160101); A47J36/06 (20060101); F24B5/06 (20060101)

U.S. Cl.:

CPC A47J37/0704 (20130101); A23L5/10 (20160801); A47J36/06 (20130101); A47J37/0763

Background/Summary

FIELD OF THE INVENTION

[0001] The present invention relates generally to BBQ air flow configurations systems in a BBQ lid. In particular, the present invention relates to control of heat zones within a BBQ by variable venting in a BBQ lid. In greater particularity, the present invention relates to methods of cooking meat on a BBQ grill by venting combustion gasses to control cooking grate temperature zones.

BACKGROUND OF THE INVENTION

[0002] A Barbecue (typical US usage is “BBQ”) grill is a cooking appliance that uses various fuels to cook meats and vegetables on a cooking grate surface. This type of cooking appliance, often referred to as “BBQ grilling” or simply “grilling,” allows for a form of cooking that involves a dry heat applied to the food, typically from below, using a variety of heating sources, such as charcoal, gas (e.g. propane or natural gas), wood chips, pellets, and often a combination of these sources. Grilling is an effective heating technique in order to cook meat or vegetables, depending upon the food and desired outcome of the cook, because a significant amount of heat can be applied in high proximity to the targeted food. However, grilling is also flexible, allowing for both rapid conventional heating, or slower indirect radiant heating of foods by altering the proximity of the heat in relation to the targeted food and through variation of venting of the utilized various fuel sources.

[0003] A standard, stand-alone BBQ grill typically consists of several parts, including a firebox, cooking grate, charcoal grate, lid, and vents. The firebox is in the main body of the grill, where the heating source, such as charcoal, is placed and lit to produce combustion and heat. The cooking grate position above the heating source is the surface on which the food is placed, while the charcoal grate is the surface on which the charcoal is placed just under the cooking grate. The lid is used to cover the grill, trapping heat and smoke to cook the food and add a savory flavor to the targeted food that is distinctive of BBQed foods. Importantly, the vents on the grill lid are used to control the airflow, which in turn controls the temperature of the grill. The resultant concentrated heat allows for the creation of excellent tasting, cooked meats, while locking in the meats juices for a self-basting, internal savory eating experience.

[0004] In standard usage, a BBQ grill is first placed on a flat, stable surface, away from any flammable materials and the lid vents opened to allow for proper airflow through and around the firebox area. Charcoal is then placed on the charcoal grate and lit, allowing it to burn until it is covered in white ash. The combustion wait time varies based on the type of fuel utilized in the firebox, but the appearance of white ash exterior on charcoal, such as charcoal briquettes, ensures that the food flavor is untainted by flammable combustion accelerant chemicals, such as lighter fluid, that may be topically applied to the combustion material or which may be embedded on the exterior of the fuel source to facilitate the rapid and easy combustion of the fuel. For example, Match Light® briquettes is a popular brand of charcoal briquette used in grilling and includes a fuel accelerant embedded within the exterior of the briquette to allow for immediate, and effortless lighting of the entire fuel source with a minimum of user firebox tending.

[0005] Once the fuel source is prepared and partially combusted, such as when ash covers charcoal exterior, the cooking grate is placed on top of the firebox and targeted food items placed on the grate for cooking. The lid to the grill is then placed over the grate, making sure the vents are open, thereby sealing the targeted food items adjacent to the fuel source and in a partially pressurized atmosphere of combustion byproducts.

[0006] BBQ grills typically include two vents that allow for the control of the airflow within the

grill and, therefore, the temperature of the grill. One vent is positioned on the underside of the kettle lower bowl portion and another is typically positioned on the upper surface of the kettle lid adjacent to a lid handle. By opening either of these vents, more air is allowed into the grill, which increases the combustion reaction of the fuel source and thereby the temperature within the enclosed, lidded area. By closing the vents, less air is allowed into the grill, which decreases the temperature, but increases the exposure of combustion byproducts, namely, fuel source smoke, to the targeted food. It is important to monitor the temperature of the grill and adjust the vents as needed to ensure the food is cooked to the desired level. Moreover, by varying the venting of the grill a skilled BBQ cook can vary the ratio of smoke byproducts to heat and to impart the exact taste, degree of doneness, and the temperature of the targeted food placed on the cooking grate. Thus, an experienced BBQ grill cook can, especially for meats such as pork and steak, achieve a fully cooked, BBQ flavored piece of meat that has from time immemorial been sought after by BBQ meat consumers.

[0007] Positioning of the fuel source within a BBQ grill is also an important consideration to a BBQ cook. For example, a fuel source such as charcoal may be positioned uniformly onto a lower fuel grate within the BBQ or it may be layered to one side of the fuel grate to allow for indirect heating, as will be further described. In addition, modern grills also may include a conical metal accessory, the accessory having open lower and upper ends, that may be positioned on the lower fuel grate that fully contains the combustion source. The conical accessory may be positioned in a narrow end up, or a narrow end down, depending upon the type of food source to be cooked and the type of cooking strategy to be used, but the accessory is typically positioned in the center of the fuel grate. Some influencers and popular BBQ cooks referred to this accessory as a “Vortex” grill accessory, probably due to the perception that combustion gasses are concentrated into a central shaft of rising heat within the grill kettle.

[0008] While vegetables are often cooked on a BBQ grill, especially in a Shish Kebab arrangement, meats are typically targeted for cooking on a BBQ grill. A variety of meats can be cooked on a BBQ grill, including beef, pork, chicken, and fish. Each type of meat requires a different cooking time and temperature, so a cook's experience is an important factor to achieve a good taste and texture to the targeted food. Hence, proper control and precise adjustment of lid vents are needed to ensure the proper preparation of the meat. Beef, for example, can be cooked to various levels of internal temperatures, from rare to well-done, while chicken should always be cooked to an internal temperature of 165° F. to ensure it is safe to eat. Fish, on the other hand, should only be cooked until it flakes easily with a fork, but not overdone such that the smoke flavoring does not overpower the delicate fish flavor.

[0009] Therefore, a BBQ grill is a versatile, popular cooking appliance that can be used to cook a variety of meats and vegetables to perfection. While a BBQ grill includes a fairly simple arrangement of grill elements, including a firebox, a firebox vent, a cooking grate, a charcoal or fuel grate, a lid, and a lid vent, the arrangement creates a controlled cooking environment that permits a skilled user to cook a variety of foods and impart a savory and highly sought after BBQ taste to targeted food. The venting on the lid is particularly important, as it allows for the control of the airflow and temperature of the overall grill.

[0010] However, typical BBQ grills only include a single vent in the flat, top portion of the lid which only permits control of the entire cooking interior of the grill. This current arrangement forces cooks to often use multiple BBQ grills simultaneously so that the cook may vary the conditions on each grill separately to create a suitable cooking environment for the quantity and type of foods that are being cooked. However, quite often, one grill is sufficiently large to accommodate all of the food that needs to be cooked so that using multiple grills, or grills used in succession, could be avoided if the cooking conditions may be varied in different areas within the single grill. This feature is not available with a BBQ grill that has a single grill vent in the upper surface of the BBQ lid.

[0011] Therefore, what is needed is an improved lid venting system that allows for the variable control of the venting gases by a user to create various cooking zones within a single BBQ grill that can accommodate different types of targeted foods within a single grill to assist a cook in the perfection of BBQ food preparation.

SUMMARY OF THE INVENTION

[0012] It is the object of the present invention to provide a variable venting system that is embedded in the lid of a BBQ grill. In particular, the venting system includes a plurality of meshed gears that permits the fine tuning of multiple venting orifices or openings into the heated interior of the grill cooking area under the lid. The variable vents are arranged in a circumferential pattern around the lower circumference of the rounded BBQ lid so that grouping of controlled temperature zones and internal atmospheric adjustments may be made to each cooking zone area via an exterior adjustment of a selected venting group. The vents also permit the promotion of a high volume of air to be exhausted from the BBQ lid thereby causing a greater amount of combustion within the cooking kettle.

[0013] Other features and objects and advantages of the present invention will become apparent from a reading of the following description as well as a study of the appended drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0014] A meshed gear venting system incorporating the features of the invention is depicted in the attached drawings which form a portion of the disclosure and wherein:

[0015] FIG. 1 is a perspective view of a typical BBQ grill;

[0016] FIG. 2 is an exploded view of the typical BBQ grill shown in FIG. 1;

[0017] FIG. 3 is a top plan view of a typical lid of the BBQ grill shown in FIG. 1 and FIG. 2;

[0018] FIG. 4 is a top plan view of an improved BBQ grill lid incorporating an improved lid venting system;

[0019] FIGS. 5A-5D show 4 perspective view of a vent used in the improved lid venting system;

[0020] FIGS. 6A and 6B shows two perspective views of a vortex accessory used in an inverted and standard fashion;

[0021] FIG. 7 is a top perspective view of a BBQ grill in which the fuel source has been sequestered to one side of the fuel source grate;

[0022] FIG. 8 is a plan view of the cooking grill element of a BBQ system showing four quadrant, cooking zones controlled by vents arranged within the BBQ cooking vessel lid; and,

[0023] FIG. 9 is a perspective diagrammatic view of the heated combustion air flow under the cooking vessel lid.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

[0024] Referring to the drawings for a better understanding of the function and structure of the invention, FIG. 1 shows a typical prior art BBQ grill 10 having typical functional components, such as a cooking kettle 11, an ash catcher 12 and a support stand 14, typically including wheels for easy relocation of the grill 10. All BBQ grills are generate large amounts of heat to lock in the juices on cooked meats, so almost all materials in the grill 10 are made of some type of metal, such as low grade steel and aluminum, that can withstand large amounts of heat.

[0025] As showing in FIG. 2, typical BBQ grill 10 includes a number of parts that in combination allow for the versatile cooking of any type of food, however BBQ's are best known for the cooking preparation of meats due to the high amount of focused heat that they generate. A kettle section 11 includes a lid 16 that is manipulated by a handle 18 for lifting and rotation thereof. Lid 16 also typically includes a lid damper or variable venting assembly 17 for controlling the exhaustion of combustion gasses and internal temperature of the kettle cooking area. The lid portion 16 is

supported by kettle lower bowl portion **22** at its lower circumferential edge. The circumferential edge acts as a seal for the lid and provides a support for a metal cooking grate **19** that is further supported by a series of circumferential support tabs or members (not shown) formed on the inside surface of lower portion **22**. Lower portion **22** also includes within its lower interior a combustion chamber or “fire-box” **25** as it is often referred to where combustion materials such as charcoal or wood chips, or a combination thereof are burned. The combustion materials are supported by lower fuel grate **21** that sits within the lower interior of portion **22** but is partially suspended above and spaced from the internal lower surface of the lower portion **22**. Lower kettle portion **22** typically also includes a kettle bowl damper **26** that allows for the intake of air for the combustion of fuel materials held by fuel grate **21** to burn, and also for the extraction of fallen ash and combusted materials after a cooking session. The damper **26** typically includes multiple paddles rotatable around a central axis of damper **26** to variably cover and uncover several vent openings formed in the bottom of kettle portion **22**, and to agitate ash collected into the bottom of kettle **22** via an integral rotation tab, thereby causing the extraction of that ash onto the ash catcher **12**. An ash catcher **12** also helps to support a stand assembly **14** to support the BBQ grill **10**, and catch ash debris collected from kettle **11** during combustion. Lower portion **22**, and thereby the entire kettle section **11**, may be transported via lifting handles **23** that are positioned on the exterior of portion **22** and which include a heat shield to allow for limited movement of kettle **11** without injury, even when the grill is still hot.

[0026] As shown, kettle section **11** is supported by stand assembly **14**, which typically includes a combination of leg struts **29**, leg end caps **33**, a wire support assembly **32**, wheels **31** and wheel retention bolts **31a**, and some connection screws and clamps **28**. Stand **14** importantly provides stability for the operation of BBQ and allows for the short-distance transportation of the grill **10** in between grilling operations. As shown, supporting wire assembly **32** connects and supports the kettle **11** between wheels **31** thereby providing important stability to the entire grill.

[0027] As is known, ash catcher **12** and lower vent **26** may be integrated into a more complex arrangement to both catch and contain ash debris in an enclosed lower vessel, along with any meat juice and oil that may drip down into the ashes during cooking. Such an arrangement, may also include an extension handle to connect with the paddles of damper and to allow for adjustment of the damper to finely tune the amount of air allowed to enter the combustion area of lower section **22**.

[0028] Referring to FIG. 3, lid **16** is typically formed into a curved, hemi-spherical shape, but is usually flattened on its top portion, thereby forming a relatively flat circular margin **40** between concentric circles **37** and **38**. Lower kettle portion **22** in some configurations includes an auxiliary side support wire **36** on the exterior of lower portion **22** that allows for the temporary support of the lid **16** so that the lid may be rotated up and temporarily placed on the side of kettle **11** and rotated back selectively for the easy tending of cooking meat within the kettle **11**. Handle assembly **18** is supported from and bolted to the inside of lid **16** via a resilient metal mesh **39**, and vent **17** is typically located at or within margin **40** as shown. As is known, vent **17** may be adjusted so that one or more apertures **41** may be selectively closed to control the interior combustion reaction in kettle **11** as described above.

[0029] Because vent **17** is in a fixed location within portion **40**, a grill cook may depending upon the location of combustible fuel held by fuel grate **21** rotate lid **16** to control the temperature in a portion of the cooking grill **19** that is directly beneath vent **17**. By rotating the lid, the changing position of the vent **17** somewhat affects the temperature on grill **19** thereby slightly changing the cooking conditions on a portion of the surface of the grill **19**. However, this method is inadequate to allow for the proper separation of exhaust gasses at a level that can cook different types of meats on a grill. Hence, even when a typical chef rotates the grill lid **16** in an attempt to adjust heating for a portion of the cooking grill, or simply to ensure that each area is thoroughly cooked, the vent assembly **17** is limited in its ability to meet the requirements of an accomplished BBQ chef, and

usually the vent assembly **17** is repeatedly manipulated with the hope that the adjustment will meet the cooking goals, but usually with only limited success.

[0030] Referring now to FIG. **4**, it may be seen that lid **16** is formed to include an improved venting system **50**. FIG. **4** shows a modified lid in a plan view with the BBQ grill lower section **22** positioned below lid **16** and not visible. As shown, venting system **50** includes four adjustable vents **51** positioned within a new margin area **40** relocated downward from the flat margin area shown in FIG. **3**. Each vent **51** is formed into a spherically contoured shape to match the curvature of the surface of relocated area **40**, now bounded by repositioned concentric circle lines **37** and **38**. Lid **16** further includes handle assembly **18** and a temperature gauge **43** for measuring temperature within kettle **11**.

[0031] Each venting zone **51** includes a plurality of venting gears **53**, as will be further discussed. However, importantly, vents **51** are positioned on the lower portion of the downward curved surface **46** of lid **16** so that combustion gases held within kettle **11** are allowed to flow forcefully outward and through the vents of the system **50**. As discussed in FIG. **3**, substantially flat zone **40** upon which vent **17** is formed, only allows for combustion gases to be forced upon and out the zone **40** on the top of the lid **16**. Conversely, vents **51** are positioned well below the flat portion **40** so that hot combustion gasses flow around and over any meat or other targeted food material for cooking. Importantly, vents **51** are formed at equal distances around new zone **40** defined by inner and out limits **37**, **38**. Zone **40** is formed, along the circumferential edge **44** of lid **16** such that vents **51** are positioned approximately 1-2 inches from lower edge **44**. As will be discussed, each vent **51** creates four independent cooking zones on the surface of cooking grill **19**.

[0032] Referring to FIGS. **5A-5D**, each vent **51** includes a plurality of venting gears **53** that are supported within an exterior shroud panel **52** with each gear centered within a circular opening **54** formed within shroud **52** on its upper, interior surface thereof. Gears **53** are held in place against the exterior surface of lid **16** with the shroud which is pinned, spot welded, or glued onto the lid surface, as may be understood. Shroud **52** is contoured to create an integrally formed compartment **55** with opening **54** to accommodate each gear **53** such that each gear **53** may freely rotate within the shroud compartment **55** in a floating arrangement biased against the exterior surface of lid **16**. The shroud is spaced away from the lid surface such that a small amount of friction is exerted against the upper surface of each gear to slightly squeeze each gear against the exterior surface of lid **16**. This arrangement prevents the free-wheeling of gears **53** so that once rotated to a desired rotational position each gear stays at that rotational setting, yet allows for adjustable movement of each gear. To achieve this precise spacing relationship, shroud **52** is preferable curved in three dimensions to follow the curved exterior surface of lid **16** as shown in the figures in order to hold each gear within its own compartment while freely allowing controlled rotation of each gear.

[0033] Each venting gear **53** includes a plurality of venting apertures **58** and each gear as described preferably is held within shroud compartment **55** in a floating arrangement. Alternatively, a center screw or pin may retain gear **53** through aperture **57** in lieu of a floating arrangement. Each vent **51** may have its gears **53** rotated via upwardly extending lever **56** located on one of the end gears due to a synchronized, meshed movement relationship. Each gear is meshed with a gear positioned adjacent to another gear as shown so that as the gear with the handle or lever **56** is rotated all gears within the vent **51** are simultaneously rotated. Lid **16** defines a set of lid apertures or holes **61** formed therethrough which are sized and positioned appropriately so that rotation of each gear may variably occlude each holes **61** depending upon the rotation angle of each gear in relation to the lid apertures **61**. Notably, apertures **58** are spaced evenly with respect to the circumference of gear **53** so that as each gear is rotated occlusion of each corresponding lid aperture **61** underneath apertures **58** occurs equally and simultaneously for each gear. As shown, each venting gear **53** is positioned to mesh with an adjacently positioned gear so that as one gear is rotated, an adjacently positioned gear is counter-rotated. The apertures **61** in lid **16** are sized and positioned appropriately so that rotation of each gear **53** may fully or partially occlude venting holes **61** depending upon the

rotation angle of each gear in relation to the holes in lid **16** irrespective of the rotation direction of each gear **53**. When the apertures in venting holes **58** are aligned with apertures **61**, combustion gases freely pass through the venting zone **51** and out of the side of lid **16**, thereby allowing for a more rapid passage of hot combustion gasses to escape from the open and aligned apertures. Gears **53** may be rotated 360 degrees so that that amount of occlusion may be varied by a user to control airflow through each vent gear **53**, and thereby flow through the entire vent **51** as may be understood. The size and availability of holes in each gear **53** are such that the combined effect of each vent **51** is at least a ten-fold increase of air flow out of a single vent, as compared to a standard vent **17**, when vent **51** is fully opened.

[0034] Referring to FIGS. **6A** and **6B**, it may be seen the use of a cone-shaped, vortex cooking accessory **101**. The accessory has large lower end **103** and a smaller, constrained opening end **102**, and is made of standard heat resistant metal formed into a conical shape as shown. The accessory is sized to hold a substantial quantity of fuel **110** within its interior and supported by fuel grate **21** such that combustion gases escape through the upper constrained end **102**. Due to the shape of the accessory **101**, combustion gases are shaped to extend upward vertically concentrating the hot gasses **107** into a vertically shaped shaft. The vortex accessory **101** is used to provide flexibility in how food is cooked on a BBQ grill and such a cooking method is often referred to in the industry as “vortex” cooking **100**. Vortex cooking in a BBQ grill is a method that uses the conical accessory **101** to create an indirect heat zone **108** around the accessory but within the grill **10**. The accessory **101** is typically, but not always, placed in the center of the grill **10** and filled with lit charcoal **110**. Usually, a second accessory (not shown) called a “chimney” is used to hold a quantity of charcoal or other fuel source that is pre-lit and combusted within the chimney until the charcoal is well lit and substantially involved with flame or is red-hot in appearance. The burning coals are then dumped into the vortex accessory **101** so that the grill **10** is ready for cooking. The heat from the charcoal in the vortex **101** is directed upwards and then radiates down around the grill, creating an even, high heat zone **108** surrounding the vortex **101**. This is referred to in the industry as “indirect” or “radiant” heating so that food **106**, such as a beef steak or chicken wings, are not exposed directly to an open combustion gas positioned directly below the food as is typical in a standard BBQ grill. This type of indirect heating, sometimes also referred to as “low and slow” cooking, produces thoroughly cooked meat with a crispy exterior and is an especially desired method for cooking chicken wings. Often times, the cooking grill **19** is shaped to accommodate a vortex type accessory so that the upper opening **102** is even with or slightly protrudes above the level of the cooking grill surface in a circumscribed area **109** formed by the cooking grill **19**.

[0035] FIG. **6B** showing the same accessory **101**, but in an inverted position having the larger end **103** above the smaller constrained end **102**. In this configuration, a cooking grill **19** is positioned over the inverted accessory **101** and the targeted food, such as a beef steak, positioned over the larger open end **103**. The larger opening **103** in this configuration creates an expanded direct heating exhaust area **105** in which exhaust gases **107** directly heat meat placed on a cooking grate **19** directly over the exhaust area **105**. This produces rapid heating of the exterior of the meat **106** to sear the exterior of the meat and lock in juices.

[0036] In another application of an inverted vortex accessory **101** depicted in FIG. **6B**, a stand sometimes referred in the industry as a “beer can” stand (not shown) is placed at the center of the inverted accessory and a whole chicken is placed over the beer can stand wholly enclosed within the inverted accessory **101**. Hot coals, such as created with a chimney as described above, are scattered around the exterior vortex accessory so that chicken is cooked again with indirect heating, but from outside of the larger end **103** of the accessory **101**. Such an indirect method of heating is preferred by many BBQ grill chefs because the stand may include various types of cooking ingredients, such as for example beer, that will vaporize and impart desired flavors on the chicken during cooking.

[0037] As will be understood, in all arrangements shown in FIGS. **6A** and **6B**, lid **16** would be

placed over the cooking grill **19** to enclose the vessel **11** during cooking. In addition, as may be further understood, the vortex accessory may be positioned underneath cooking grill **19** in various locations to further control the strength of radiant heating or direct heating in one direction or another.

[0038] A further indirect method **115** of cooking is shown in FIG. 7. Coals **116** as a fuel source are placed to one side of the fuel grate **21** and a targeted meat **117** is placed on the opposite side of the kettle **11** on the cooking grate **19**. Customarily, lid **16** is simply placed over the lower vessel **22** to form a closed vessel **11** during cooking. However, in this cooking method **115** lid **16** is slightly repositioned or displaced in the direction of the targeted meat **117** so that a slight gap is created just to the inside of the upper lip of vessel **22** adjacent to the meat **117**. This type of cooking creates a drafting effect such that combustion gasses **107** are drawn toward the meat **117** thereby cooking the meat **117** more quickly and with a seared outside. However, with multiple zones, heat is not contained within cooking kettle **11** as efficiently as with an enclosed lid, and other cooking zones as will be discussed suffer a loss of cooking heat, thereby extending the cooking time of some foods on the grill **19**.

[0039] Now referring to FIG. 8, cooking grate **19** is shown with the lower lid zone **40** of FIG. 4 superimposed onto the grate **19** in phantom to form a circular zone **73** that represents the zone into which vents **51** would be positioned. The cooking grate **19** is separated into a plurality of cooking zones **70(A-D)** that form, for example, four quadrants of temperature and atmospheric conditions separated along lines **71**. Each quadrant **70** includes a single venting zone **60** that together forms an entire venting system **50**. The four quadrants A-D intersect at the center **75** of cooking grate **19** and extend outward to the furthest extent of the interior of the kettle **11** under lid **16**, providing four separate cooking zones that are adaptable to the type of meat placed in the respective zone. As may be understood, each venting zone may be adjusted independently of any other zone by rotating the venting gears **53** as selectively adjusted by a user via handle or lever **56**. Hence, as each zone **60** is adjusted, the temperature and atmospheric conditions in any respective cooking zone **70(A-D)** may be altered as needed. This allows for a user (i.e. a grill cook) to sequester different types of foods, such as beef, poultry, vegetables or fish, into specific zones **70** and adjust the venting zone conditions **60** to selectively create a cooking condition (e.g. temperature) that suits the cook's taste preferences for that type of meat. This also allows for the simultaneous cooking of various foods on a single cooking surface within a single kettle **11**.

[0040] For example, cooking zone **70A** may hold beef steaks that the user wishes to heat to the maximum extent possible and expose the beef to a high level of combustion products to thoroughly caramelize the external fat of the meat, thereby adding flavor to the meat per the heating caused by the particular selected combustion materials. For that maximum heating condition, the venting gears may be rotated to allow for a full-flow of combustion gases to flow into zone A, thereby providing extra heat to that particular zone. Conversely, zone C may hold fish and the user may wish to partially or fully close the venting zone above zone C so that combustion in that zone is somewhat retarded, but still allowing for full cooking. Other zones may require partial or fractional closure of its superior vent zone to best suit the food in their respective zones. Hence, by varying each zone vent, a user can vary the cooking conditions in each zone to best suit the food in that zone and cook all foods simultaneously, but within somewhat different atmospheric grilling conditions.

[0041] The above described cooking zones A, B, C and D are also desirable in a vortex cooking configuration (see FIGS. 6A-6B) because a smaller proportion of fuel (vs standard cooking fuel distribution) is positioned directly and contained in the center of the grill whereas the food source is then placed around the circumference of the circular fuel source. This setup affords the ability for either extreme or low temperature (without direct convection) for superior results. The vortex may also be moved to the edge of the kettle and used in combination with the current venting system to enhance the vortex cooking effect for a particular zone of directed vortex heating.

[0042] Further air dynamics created by the improved venting system **50** are shown in FIG. **9**. FIG. **9** shows a sequestered pile of charcoal **121** positioned on fuel grate **21** on a side opposite to a targeted piece of food, such as a steak **122**. Vent **51a** is positioned adjacent to steak **122** and gears **53** rotated to open vent **51a** fully to create a hot cooking zone **60** conducive for a selected condition desired to cook a steak. In operation, as vent **51a** is adjusted, air is vented and flows along exhaust airflow path **124** to surround and pass close by meat **122**. This occurs with lid **16** fully seated over lower vessel **22** thereby holding the full exhaust heat within cooking vessel **11**, thereby cooking and searing meat **122** rapidly, and allowing for other foods in other cooking zones **60** to be efficiently cooked, but forcing a concentrated amount of heat on the steak in zone **60**.

[0043] While FIG. **9** shows a sequestered arrangement of charcoal, it will be understood that a full bed of coals can similarly be arranged on fuel grate **21** and vents **51** adjusted so that different cooking conditions are created as is depicted in FIG. **9** by varying the degree that each vent **51** is opened. Similarly, a vortex accessory (see FIG. **6A**) may be utilized and vents **51** adjusted so that the exhaust cases are directed toward a particular cooking zone. For example, 3 of the 4 cooking zones **60** may be loaded with chicken wings, but a 4th zone loaded with a piece of steak. The vent **51** adjacent to the steak may then be adjusted to draw exhaust gasses from the top of the vortex accessory over and around the steak so that while the chicken wings are cooked with radiant or indirect heat, the steak is cooked with direct exhaust gases to sear the exterior of the steak. Hence, a skilled cook may vary each vent **51** to adjust the cooking conditions of each adjacent cooking zone **60** allowing for the efficient and variable cooking of foods placed on cooking grate **19** with the lid **16** fully enclosing kettle **11**.

[0044] While I have shown my invention in one form, it will be obvious to those skilled in the art that it is not so limited but is susceptible of various changes and modifications without departing from the spirit thereof.

Claims

1. A BBQ venting system, comprising: a. a BBQ lid seated upon and enclosing a combustion firebox for a BBQ grill, said firebox including an upwardly extending vessel above and surrounding said firebox for cooking food, said vessel supporting a cooking grate in said food cooking area, said lid further having a substantially hemispherical shape, and wherein said lid is seated upon said upwardly extending vessel at an upper edge thereof thereby sealing said food cooking area; b. at least one vent formed along a lower curved exterior of said lid, wherein said vent is positioned at a level just above and adjacent to said food cooking area and providing egress of hot cooking gases from within said food cooking area; c. said vent including a plurality of meshed gears positioned adjacent to one another and each gear having a plurality of venting apertures formed therein; and, d. wherein said lid further includes a plurality of grouped apertures positioned cooperatively under each said gear in said vent such that as each said gear is rotated said apertures on each said vent are rotationally aligned with said lid grouped apertures to permit variable exhaust of combustion gases held within said lid during cooking.
2. The venting system of claim 1, wherein each said vent further comprises a shroud holding a series of said meshed gears biased against said lid while allowing substantially free rotation thereon.
3. The venting system of claim 2, wherein said vent is three-dimensionally contoured to match the hemispherical surface of said lid at a lower portion thereof.
4. The venting system of claim 3, wherein each said vent defines a cooking zone on said cooking grate such that varying said gears controls a cooking condition in said cooking zone.
5. The venting system of claim 4, wherein said venting system comprises four cooking zones under said lid defined by a single vent in said lid for each cooking zone.
6. The venting system of claim 5, wherein said system includes a vortex accessory positioned in

said firebox and said at least one vent draws combustion gases from an annular upper area of said vortex accessory over and around food placed on said cooking grate adjacent to said vent.

7. The venting system of claim 5, wherein said venting system includes a vortex accessory in said firebox and a plurality of vents are arranged around the lower periphery of said lid such that indirect heating of the interior of said lid is uniformly maintained thereby obviating the necessity of rotating said lid during cooking.

8. The venting system of claim 1, wherein at least one of said gears includes a lever extending outward from said gear for rotation of said same.

9. The venting system of claim 8, wherein said gears are arranged in counter rotating fashion relative to one another.

10. The venting system of claim 9, wherein said vent is three-dimensionally contoured to match the hemispherical surface of said lid at a lower portion thereof.

11. The venting system of claim 2, wherein said contact point between said lid and said firebox defines a circle, wherein said venting system comprises a plurality of vents wherein said vents are positioned circumferentially with respect to said circular contact point, and wherein said vents are equally spaced from one another and on opposite sides of said circular contact point.

12. The venting system of claim 1, wherein said contact point between said lid and said firebox defines a circle, wherein said venting system comprises a plurality of vents wherein said vents are positioned circumferentially with respect to said circular contact point, and wherein said vents are equally spaced from one another and on opposite sides of said circular contact point.

13. In a BBQ grill having a stand supporting a substantially sealed lower portion, said substantially sealed lower portion forming a fire box portion contained within a lower area of said lower portion, a cooking rack supported by said lower portion for holding food items for cooking, fuel contained within said fire box portion for supplying heat to cook said food items, and a lid supported by said lower sealed portion for sealing fuel caused heat within said BBQ grill, an improved venting system comprising: a. a plurality of vents arranged around the circumference of said lid at a lower portion thereof adjacent to said lower sealed portion; b. said lid defining a series of apertures collocated and beneath each said vent; c. wherein each said vent includes a plurality of meshed gears for causing rotation of all said gears upon the rotation of any one of said gears; and, d. each said gear defining a plurality of openings aligned with said lid apertures and cooperatively spaced such that rotation of any gear selectively provides a variable passageway to exhaust combustion gases from burning said fuel in said fire box.

14. The venting system of claim 13, wherein each said vent defines a cooking zone on said cooking rack such that varying said gears controls a cooking condition in said cooking zone.

15. The venting system of claim 14, wherein said venting system comprises four cooking zones under said lid defined by a single vent in said lid for each cooking zone.

16. The venting system of claim 15, wherein at least one of said gears includes a lever extending outward from said gear for rotation of said same.

17. The venting system of claim 13, wherein each said vent further comprises a shroud holding said meshed gears biased against said lid while allowing substantially free rotation thereon;

18. The venting system of claim 17, wherein said contact point between said lid and said firebox defines a circle, and wherein said vents are positioned circumferentially with respect to said circular contact point and are equally spaced from one another and on opposite sides of said circular contact point.

19. The venting system of claim 13, wherein said system includes a vortex accessory positioned in said firebox and said vents draw combustion gases from an annular upper area of said vortex accessory over and around food placed on said cooking rack adjacent to each said vent.

20. A method for improved cooking in a BBQ grill, comprising the steps of: a. lighting a fuel source within said grill to create combustion gases under a lid sealing said combustion gases within said grill; b. placing different types of meats in one or more cooking zones on a cooking grate held

under said sealing lid within said grill; c. adjusting vents adjacent to each cooking zone to control the temperature within each said zone; d. wherein said vent adjusting step comprises the steps of: i. selectively rotating a plurality of meshed gears to allow hot combustion gases to pass through said lid at said vent location; ii. wherein said gear rotation step adjusts the amount of gases that pass through said vent dependent upon the degree of rotation selected for said meshed gears; and, iii. wherein said adjustment step causes combustion gases to flow over and around food supported by said cooking grate positioned adjacent to an adjusted vent.

21. The method of claim 20, wherein said step of placing meats in said one or more cooking zones comprises the step of placing said meats in equally spaced zones around said cooking grate, wherein each cooking zone and said vents are spaced at equal intervals from one another around the exterior of said BBQ grill.

22. The method of claim 21, wherein said step of lighting a fuel source further comprises the step of lighting a fuel source held within a vortex accessory, and wherein said vent adjustment step cause combustion gases exiting said vortex accessory to flow towards a selected adjusted vent.

23. The method of claim 21, wherein said step of lighting a fuel source further comprises the step of lighting a fuel source contained within a vortex accessory placed at the center of said BBQ grill, said improved cooking method further including the step of adjusting each vent such that combustion gases flow equally out of each vent thereby uniformly and indirectly heating the interior of said BBQ grill.
